Research Topics / Open Problems

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Topic 1: Index Coding

Recent Activity and Research

- A single sender broadcasts messages to many receivers; each receiver already has some messages as "side information" and wants one it lacks. The question studied is how few transmissions the sender needs so that everyone gets what they want.
- Introduced by **Birk & Kol (1998)** and formalized as a graph problem by **Bar-Yossef, Birk, Jayram & Kol**, who tied the optimal number of transmissions to a graph parameter called **minrank**.
- **Golovnev, Regev & Weinstein** prove tight bounds on the minrank of random graphs; a related result by **Chawin & Haviv (2024)** shows minrank is NP-hard to approximate within any constant factor.
- **Haviv (2025)** gives linear-size kernels for the Storage Capacity and Dual Index Coding problems.
- **Blasiak, Kleinberg & Lubetzky** use a linear-programming hierarchy to bound the broadcast rate, and **Lubetzky & Stav** show that nonlinear index codes can strictly beat the best linear ones – keeping the search for tighter bounds and better schemes an active problem today.

Subtopics to be Studied

- **Problem setup and graph model** – the broadcast-with-side-information problem, and its representation as a "side-information graph."
- **Coding schemes** – clique-cover based schemes and the graph parameters that measure their performance.
- **Minrank** – the linear-algebraic graph parameter that exactly characterizes the best possible *linear* index code, and how to compute/bound it.

- **Converse techniques** – outer bounds such as the maximum acyclic induced subgraph (MAIS) bound, used to show a scheme cannot be improved.
- **Vector-linear codes** – using vectors instead of single symbols per message, and why this can beat scalar-linear schemes.
- **Nonlinear index coding** – why linear codes are not always optimal, and constructions that beat the linear optimum.
- **Related structures** – connections between index coding, network coding, and matroid theory.

Study Resources

- F. Arbabjolfaei, Y.-H. Kim, ***Fundamentals of Index Coding***, Foundations & Trends in Comm. and Inf. Theory, 2018 (main reference). https://ieeexplore.ieee.org/document/8497038/
- Y. Birk, T. Kol, "Informed-source coding-on-demand (ISCOD) over broadcast channels," INFOCOM 1998. https://ieeexplore.ieee.org/document/662940/
- Z. Bar-Yossef, Y. Birk, T. S. Jayram, T. Kol, "Index Coding with Side Information," FOCS 2006. https://webee.technion.ac.il/people/zivby/papers/indexing/indexing.focs.pdf
- A. Golovnev, O. Regev, A. Weinstein, "The Minrank of Random Graphs." https://arxiv.org/html/1607.04842
- D. Chawin, I. Haviv, "Improved NP-Hardness of Approximation for Orthogonality Dimension and Minrank," SIAM J. Discrete Math., 2024. https://arxiv.org/abs/2301.00732
- I. Haviv, "Kernels for Storage Capacity and Dual Index Coding," 2025. https://arxiv.org/abs/2504.12274
- J. Blasiak, R. Kleinberg, E. Lubetzky, "Index coding via linear programming." https://arxiv.org/pdf/1004.1379
- E. Lubetzky, U. Stav, "Nonlinear index coding outperforming the linear optimum." https://ieeexplore.ieee.org/document/5165171/

Topic 2: Entropy Region and Non-Shannon-Type Information Inequalities

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Recent Activity and Research

- Every set of n random variables has an **entropy vector** – the joint entropies of all its subsets. The **entropy region** is the set of all vectors that can actually arise this way.
- The basic (**Shannon-type**) inequalities carve out a region containing the true entropy region, but **Zhang & Yeung (1998)** showed this containment is strict: some inequalities that entropy vectors must obey cannot be derived from Shannon-type inequalities alone.
- This connects to matroid theory through the **Ingleton inequality**, and to group theory through **group-characterizable entropy functions**, studied by **Chan & Yeung**.
- **Matveev & Romashchenko (2026)** study the structural properties and stability of the Ingleton inequality, and **He & Chen (2026)** classify entropy functions on low-dimensional faces of the polymatroid region.
- **Csirmaz (2025)** surveys known methods for deriving new entropy inequalities, including the Copy Lemma technique.
- Fully describing the entropy region for four or more variables remains an **open problem**.

Subtopics to be Studied

- **Entropy functions and polymatroids** – how entropy vectors satisfy polymatroid axioms, giving the basic Shannon-type outer bound.
- **The I-Measure** – a set-theoretic (Venn-diagram-like) way of visualizing entropy and mutual information, useful for manipulating entropy identities.
- **Non-Shannon-type inequalities** – the Zhang–Yeung inequality as the first known example beyond Shannon-type inequalities, and how such inequalities are found.
- **Ingleton inequality and matroids** – how matroids give rise to entropy-like functions, and how matroids violating Ingleton (e.g., the Vámos matroid) relate to the entropy region.
- **Group-characterizable entropy functions** – the Chan–Yeung link between entropy functions and finite groups, an algebraic route to generating entropy vectors.
- **Geometry of the entropy region** – extreme rays and faces of the region, and how tightly known inequalities pin it down for four or more variables.

Study Resources

- R. W. Yeung, ***Information Theory and Network Coding***, Springer, 2008 (main reference; Ch. 6, 14, 15, 16). https://iest2.ie.cuhk.edu.hk/~whyeung/post/draft2.pdf
- L. Csirmaz, "Exploring the entropic region," 2025. https://arxiv.org/html/2509.12439
- S. Matveev, A. Romashchenko, "Structural Properties of Entropic Vectors and Stability of the Ingleton Inequality," 2026. https://arxiv.org/pdf/2512.02767
- K. He, Q. Chen, "Entropy Functions on Two-Dimensional Faces of Polymatroid Region with One Extreme Ray Containing Rank-One Matroid," 2026. https://arxiv.org/abs/2602.03363